1. The process of converting from plaintext to cipher text is known as deciphering or decryption.
   a. Yes
   b. No
2. If both sender and receiver use the same key to encrypt and decrypt, the system is referred to as:
   a. Public-key crypto system
   b. Asymmetric key crypto system
   c. Symmetric key crypto system
   d. None of the above
3. When message is divided into blocks (e.g., sets of 8- or 16-bit blocks) and each is transformed into encrypted block of cipher using algorithm and key it is called
   a. Block Cipher
   b. Bit Stream Cipher
   c. Both a and b
   d. None of the above

4. Which of the following is a symmetric crypto system using 64-bit blocks?
   a. RSA
   b. DES
   c. PGP
   d. None of the above
5. What size key does a DES system use?
   a. 64 bit
   b. 128 bit
   c. 56 bit
   d. 256 bit
6. What is data encryption standard (DES)?
   a. Block cipher
   b. Stream cipher
   c. Bit cipher
   d. None of the above

7. Which is the largest disadvantage of the symmetric Encryption?
   a. More complex and therefore more time-consuming calculations
   b. Problem of the secure transmission of the Secret Key
   c. Less secure encryption function
   d. Isn't used any more
8. What you should be most careful of when looking for an encryption method to use?
   a. Complexity of the algorithm
   b. Veracity of the vendor’s claims
   c. Speed of the algorithm
   d. How long the algorithm has been around
9. Which of the following is most likely to be true of an encryption method that is advertised as unbreakable?
   a. It is probably suitable for military use.
   b. It may be too expensive for your organization.
   c. It is likely to be exaggerated.
   d. It is probably one you want to use.

10. Which of the following is most true regarding certified encryption methods?

a. These are the only methods you should use.

b. It depends on the level of certification.

c. It depends on the source of the certification.

d. There is no such thing as certified encryption.

11. Which of the following is most true regarding new encryption methods?

a. Never use them until they have been proven.

b. You can use them, but you must be cautious.

c. Only use them if they are certified.

d. Only use them if they are rated unbreakable.

12. Asymmetric encryption utilizes only a public key for encryption and decryption.

a. True

b. False

13. Asymmetric encryption can be used for confidentiality but not for authentication.

a. True

b. False

14. Public-key encryption is more secure than symmetric encryption as it uses two keys.

a. True

b. False

15. The most widely used public-key cryptosystem is _____ .

a. RSA

b. DES

c. 3DES

d. None of the above

16. The Diffie-Hellman algorithm depends on the difficulty of computing discrete logarithms for its effectiveness.

a. True

b. False

17. Most of the products and standards that use public-key cryptography for encryption and digital signatures use RSA.

a. True

b. False

18. Suppose that you are using the RSA algorithm and have chosen two prime numbers, p = 5 and q = 11. What will be value of Phi

a. 55
b. 13
c. 14
d. None of the above

19. In RSA algorithm, if the two prime numbers are, 5 and 11 then what will be value of Modulus, N

a. 55
b. 44
c. 20
d. None of the above

20. Given two very large prime numbers p and q, a composite number N (Modulus) can be calculated as N = p* q. However, given just N, it is not feasible for hackers to find p and q.

a. Yes, that is true
b. No, that is not true at all
c. It may be possible in some cases
d. None of the above

21. In public key cryptosystem when Carol sends a message to Bob for confidentiality, what key will she use to encrypt the message?

a. Public Key

b. Bob's Public Key

c. Carols' Private Key

d. None of the above.

22. In public key cryptosystem when Carol sends an encrypted message to Bob for confidentiality, what key will Bob use to decrypt the message?

a. Bob's Private Key

b. Bob's Public Key

c. Carols' Private Key

d. Carol's Public Key

23. In a classroom with 30 students and a professor, how many public and private keys will be used if all students want to submit the encrypted assignment to the professor?

a. 30 public and 30 private

b. 1 public and 30 private

c. 1 public and 1 private

d. Variable keys depending on the application

24. In asymmetric cryptography system for confidential communication, the private key is used by ---------- to decrypt ciphertext

a. Sender
b. Receiver
c. Sender and receiver
d. All the connected devices on the network

25. A _____ is a digital document issued and digitally signed by Certification Authority that binds the name of a subscriber to a public key.

a. Digital certificate
b. Digital signature
c. Both a and b
d. None of the above

26. The principal object of a hash function is _____ .

a. Data integrity
b. Authentication
c. Both a and b
d. None of the above

27. The most important and widely used family of cryptographic hash functions is the _____ family

a. Secure Hash Algorithm (SHA)
b. Digest
c. Hash Value
d. All of the above

28. The hash value of a message in _____ application is encrypted with a sender's private key.

a. Digital signature
b. Digital certificate
c. Secure Hash Algorithm (SHA)
d. None of the above

29. A digital signature can guarantee

a. The integrity of the message
b. The source of the message
c. Both a and b
d. None of the above

30. When a message and a digital signature (signed hash) are received, a new hash is created from the received message and compared to the original hash in the digital signature. The message has not been altered when the hash values are:

a. Not equal
b. Equal
c. It doesn't matter
d. None of the above

31. Digital signatures are created using a public key crypto system. Which key does a sender use when they wish to digitally sign a hash before it is sent?

a. Senders public key
b. Senders private key
c. Both a and b
d. None of the above

32. The purpose of a digital certificate is to -------

a. Authenticate the public key
b. Authenticate the private key
c. Both a and b
d. None of the above

33. A special mathematical function that performs one-way encryption is called

a. Asymmetric encryption
b. Transposition cipher
c. Hashing
d. Multiple encryption

34. Hybrid system uses public key cryptography for

a. Encryption
b. Key exchange
c. Both a and b
d. None of the above

35. In hybrid system ------- crypto system is used to encrypt the session data

a. Symmetric
b. Asymmetric
c. Both a and B
d. None of the above